	Azuresky	
	Management system requirements	Document no: ASDP001
Compiled by:	Sifiso Nyambi - Laboratory Manager	
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Azuresky (Pty) Ltd is entering into a coal laboratory testing market by offering more option/ marketplace of proficiency testing scheme and producer of certified reference materials. The company is looking to improve the service offering and stay on top of the competition to ensure customer satisfaction.

The service offering are as follows:

- Proficiency Testing Scheme that is designed and operated in multiple rounds for a coal testing, measurement, calibration or inspection.
- Certified Reference Material production that is prepared and tested to meet homogeneity and stability requirements.

The purpose of management system development

It is to identify, design and plan the activities which directly affect the validity of the proficiency testing scheme and certified reference material and ensure that they are carried out consistent with requirements of ISO 17043:2023; ISO 17034:2016 and ISO 17025:2017.

It is imperative that significant changes to the initial/ existing design plan which can affect validity of the proficiency testing scheme and certified material production are identified, ensure to identify the risks associated with the change, analyse, mitigate the causes to eliminate or reduce- impact thereof. Such activities may include:

- Proficiency testing item production
- Assessment of homogeneity and stability testing
- Determination of the assigned value
- Statistical analysis, etc

1. General requirements

1.1 Impartiality

ISO 17043:2023 clause 4.1, ISO 17034:2016 clause 4.2 and ISO 17025:2017 clause 4.1 require that the activities in the organisation are undertaken impartially.
The organisation has structured to manage impartiality in developing three documents:

- Risk assessment to impartiality
- Impartiality policy
- Declaration of interest

Risk assessment to impartiality document number ASRI 4.2-001 was developed, utilised to identify the risks associated with activities of running the proficiency testing scheme and reference material production. The process involved identifying risks considering potential (future) events which may negatively impact on individuals, assets, and/or the environment because of those risks, their likelihood and consequences, and actions which can mitigate these effects.

Impartiality policy document no. ASMF 4.2 was developed to formally document and outline the standards and expectations within an organization, providing clear guidance for employee behavior and decision-making. It helps eliminate micromanagement by setting clear directions, improving efficiency, and reducing stress in human resources management. Well-defined policies promote professionalism both internally, fostering a respectful work environment, and externally, enhancing the organization's reputation and business relationships.

Declaration of interest document no. ASMF 4.2.1 was developed to promoting transparency and good governance in the organisation. Again, to ensure that decisions are made in the best interests of the company and that any potential conflicts of interest are identified and managed appropriately. All team members are obligated to complete and sign the document.

1.2 Confidentiality

ISO 17043:2023 clause 4.2, ISO 17034:2016 clause 4.3 and ISO 17025:2017 clause 4.2 require that the organisation shall be responsible, through legally enforceable agreements for the management of all information obtained or created during the performance of proficiency testing scheme and reference material production.
The organisation has structured to manage confidentiality in developing three documents:

- Quality Policy Statement document no. ASMS Doc 8.2-01
- Management of confidential information document no. ASMF 4.3
- Non-disclosure agreement document no. ASMF 4.3.1 was designed to be signed by all personnel involved in proficiency testing scheme and reference material production.

1.3 Structural requirements

Azuresky Propriety Limited is registered with Companies and Intellectual Property Commission (CIPC), the registration number is 2024/223777/07. The business is located in address Unit 3, 17 Makriel Street, Wadeville, GERMISTON 1428.

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Azuresky PT scheme source coal, prepares and test as per the guidelines of ISO 17025:2017, then package and ships it for activities of the PT scheme as per the guidelines of ISO 17043:2023 and produces the reference material as guided by ISO 17034:2016. Therefore the business operates as per the guideline of three ISO standards.

There are three positions designed and defined in the management structure/organogram that are responsible and accountable for the running of the PT scheme and reference material production. It is the position of a Laboratory Supervisor, Laboratory Manager and Managing Director.

Azuresky laboratory facility conforms and participates in Azuresky Pty LTD PT scheme, the participation is on testing methods for coal which are:

- Inherent Moisture analysis
- Volatile matter analysis
- Quick Ash analysis
- Calorific Value analysis

The organisation does have other personnel as presented on the organogram who carry out their duties which include:

- In-house maintenance (eg. Maintenance of CV bombs), improvement of management system. These are Laboratory Analysts and Laboratory operators
- Except through internal audit process, all personnel are responsible to identify the deviations from the management system or from the procedures and initiate actions to prevent or minimise such deviations
- They are required to report to management on the performance of the management system.
- The task of performing internal audit is allocated to be performed by an external auditor or MD of the company.

During the development, implementation of the management system, it was important to maintain the communication of the currently developing management system in the form of training records, ensuring the participation of all personnel in the determination of actions to risks and opportunities of the management system (clause 8.5 of ISO 17043 and ISO 17025; and clause 8.8 of the ISO 17034). This approach brings the awareness to all personnel of all processes contributing to the PT and RM activities. Attendance register of the meeting are required to be maintained.

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All personnel involved in the PT activities and reference material production have access to parts of the manage system documentation and related information that are applicable to their responsibilities. All information is accessible from the server QMTEK, stored in respective folders. Permissions have been applied to restricted folders eg, HR folder.

2. Resource Requirements

This is as per ISO 17043, ISO 17034 and ISO 17025 clause 6

2.1 Personnel

The business aimed to start productively and efficiently; it was intentional to search and build a management team that is skilled in coal analysis testing methods (personnel who worked at the coal laboratory and testing coal) and who have obtained qualification in Analytical Chemistry or Chemical Engineering or Geology.

Recruiting the right people not only has a positive impact on productivity and performance but also helps reduce costs associated with turnover and poor performance. The calibre of such personnel train new employees and manage potential disruptions to operations. Additionally, high-performing employees deliver a strong return on investment (ROI)

The composition of team was characterised as follows in terms of education/qualification:

- Laboratory Manager /Laboratory Supervisor: Post graduate qualification in Analytical Chemistry or Chemical Engineering or Geology
- Laboratory Analyst: Undergraduate qualification in Analytical Chemistry or Chemical Engineering or Geology
- Laboratory Operator: Grade 12 qualification.
- Bookkeeping + payroll function: obtain external services

The organogram of the organisation is graphically presented in Document no. ASMF 5.1. All functions in the organogram have job descriptions established and implemented. It is a requirement that personnel occupying the position shall sign the document and contract of employment. This is communication to personnel of their duties, responsibilities and authorities.

- **Managing Director**: Job description title Managing Director document no. ASHR 6.2.2 E lists the responsibilities associated with the post.
- **Laboratory Manager**: Job description title Laboratory Manager document no. ASHR 6.2.2 D lists the responsibilities associated with the post.
- **Laboratory Supervisor**: Job description title Laboratory Supervisor document no. ASHR 6.2.2 C lists the responsibilities associated with the post.
- **Laboratory Analyst**: Job description title Laboratory Analyst document no. ASHR 6.2.2 B lists the responsibilities associated with the post.

- **Laboratory Operator**: Job description title Laboratory Operator document no. ASHR 6.2.2 A lists the responsibilities associated with the post.

The action of selection of personnel in the organisation shall be addressed to the requirements of stated in the procedure for the recruitment of personnel, document ASHR003. This procedure also addresses the requirements of obtaining operators/analyst through the learnership programme. These are undergraduate students at tertiary institutions.

The organisation has developed and retained records of monitoring of competence of personnel. All competence and authorizations must be reviewed annually or when there are any role changes. Records must be kept for 2 accreditation cycle. Procedure for the declaration and verification of personnel competence ASMF6.1 has also stated period/condition for the review of competence period for the retention of records. Personnel declaration form ASMF6.1.1 shall be utilized to declare personnel.

NB: Supervisor/ Laboratory Manager/ Operations Manager shall provide training to personnel in the organisation. If necessary, consultant may be requested to provide training and this will be conducted in compliance to the conditions stated for Externally provided services of the standard.

The authority to plan PT scheme and RM production is a function of the following positions of the organogram:

- Managing Director/ Operations Manager
- Laboratory Manager
- Laboratory Supervisor

The assessment of data to determine the stability and homogeneity, assigned values and associated uncertainties of the properties or PT item, evaluate the performance of the PT participants and give opinions and interpretations as well as advice to the participants is performed by Laboratory Manager and Laboratory Supervisor.

The review and to authorise the information in the form of a PT report is the Operations Manager function.

2.2 Facilities and environmental conditions

The scope of the operation as defined as sample preparation of coal samples, analysis, packaging and storage. The available and allocated space of operation was planned and partitioned, demarcated to include sample preparation area, room housing furnaces, room housing CV machine and Inherent moisture oven and a storage room for finished products (PT samples and certified reference material).
Factors affecting the validity of results such temperature control etc, were defined and have been documented in procedure titled Environmental control with the registration number ASMF 7.2.3. This is to monitor, control and record the environmental conditions.

The minimum signage of safety requirements with regards to personal protective equipment were defined and affixed to the wall. This is as per the scope of operation.

2.3 Externally provided products and services

Azuresky (Pty) Ltd seeks to enhance the production of its activities and obtain externally provided products and services. As guided from ISO 17025:2017 clause 6.6 and 7.1, ISO 17034 clause 6.2 and ISO 17043 clause 6.4, it is the company's intention to incorporate externally provided products and services into laboratory's activities. They are intended to support the operation of the laboratory.

They include but not limited to the following products and services:

- Measurement standards eg Benzoic acid, etc
- Service and repair of equipment
- Mass and Temperature calibration of equipment
- Laboratory information management system
- Consumable materials, equipment, supply of equipment components
- Assessment and auditing services
- Manpower: a service level agreement has been entered into between Azuresky (Pty) Ltd and Hlogirati (Pty) Ltd for the provision of 4 (four) undergraduate students.

NB: The condition for acquiring the service of mass and temperature calibration of equipment and service and repair of equipment competent (not necessarily accredited) service providers are authorised to be used to provide the service.

The checklist titled "Provision of equipment, services and supplies" number ASMF6.3.1 was developed and shall be used to actively monitor, measure, and managing the performance of a company's supply's to ensure they meet the required quality, cost, delivery, and other

performance expectations. As performance by suppliers can directly affect the overall efficiency, quality of products, and customer satisfaction of an organization. Furthermore, excessive slippage situations such as missed deadlines shall be flagged earlier than expected before things get out of hand allowing corrective measures to be taken more quickly! An early warning sheet ASMF 6.2 B is to be used to record such.

It's not all about the price. A company may receive invoices from suppliers following the agreed price, but unless a service or product meets expectations it is unlikely that the company can achieve savings target. It is a requirement that participants shall be informed, in advance and in writing, of products and services that are or can be provided externally, when they affect the validity of the PT activities.

Externally provided products and services procedure ASMF 6.6 has defined the purchasing requirements to be complied with. Document ASMF 6.3 is the policy required to be complied with for the provision of equipment, services and supplies.

3. Process requirements

3.1 Establishing, contracting and communicating the PT scheme objectives

The procedure to handle contractual matters, requests and tenders was developed and implemented. The document number is ASMF 4.7.1. It seeks to address all requirements of ISO 17043:2023 clause 7.1.1 and ISO 17034:2016 clause 4.1.1

There are no parts of the running of the PT scheme and reference material production undertaken by subcontractors. All parts are performed at the company.
AS026TF Outsourced test form was developed and implemented to be used for the purpose of subcontracting tests.
ASMF 6.6 Externally provided products and services procedure was developed and implemented for use of acquiring external products and services.

3.2 PT scheme communication

Azuresky PT scheme roll out plan document number ASROP 001 version 00 was developed and established to guide, inform and carry out activities to introduce the business in the market/participants. The communication of the business to participants was structured to communicate:

- the company profile (hardcopy) document
- AS002PT, Application for proficiency testing participation form. It addressed all requirements stipulated in ISO 17043:2023 clause 7.1.2.1 which are the objectives of the PT scheme, criteria to participate, criteria for determining the assigned values and evaluation, etc. It was shared with clients, completed and received back.

Company website www.azuresky.it.com communicates the business profile including objectives, complaints and appeals procedure to be followed, other services, etc.

The business shall communicate any changes in the PT scheme design or operation timely and maintain records to that effect. This is to avoid service impairment and to ensure efficient participation.

3.3 Equipment

The laboratory developed the coal testing methods of proximate analysis, the determination of calorific value and sample preparation procedure. Factors that are required for the correct

performance of PT and Reference material activities and that can influence the results are defined and addressed in each respective coal testing method.
Therefore all requirements for measurement procedure necessary for each testing method are described in compliance to the following listed clauses:

- ISO 17025 clause 6.4 and 6.5
- ISO 17034 clause 7.7; 7.8 and 7.9

Clause 7.2.2 ISO 17025: 2017 method validation of the coal testing methods were performed for the following lists of test methods.

- Inherent Moisture analysis
- Volatile matter analysis
- Quick Ash analysis
- Calorific Value analysis

A combination of the following techniques were used:

- Calibration or evaluation of bias and precision using reference standards
- Systematic assessment of factors influencing the result
- Testing method robustness through variation of controlled parameters
- Evaluation of measurement uncertainty of the results based on an understanding of the theoretical principles of the method and practical experience of the performance of the test method.

A record of the method validation per test method has a conclusion statement detailing its fitness for the intended use.

It is important to note that the section of Production Planning requirements clause 7.2 described in ISO 17034, clause 7.3 production control, clause 7.4 Material handling and storage and clause 7.5 material processing are described and addressed in each coal testing method (Quick Ash, CV, VM and IM) and coal sample preparation procedure. However, the organisation described, established procedure discussing each section of process requirements. The table below informs of the title of procedures developed aligned with the applicable requirement from the standard.

Procedure	Aim	Activities	International standard requirement
ASMF7.2.1 Production Planning	The goal of the production plan is to design the	• Review client requests and market need to identify the specific	**ISO 17034:2016 clause 7.2**

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Procedure	Aim	Activities	International standard requirement
	workflow that minimises wastage, attains PT and RM samples that have achieved sufficient homogeneity and stability requirements.	coal qualities (e.g., thermal coal, metallurgical coal, high/low ash, different calorific values). • Document the intended use of the material (PT scheme, CRM, Calibration or internal QC). • If possible, conduct a feasibility assessment (enquiry) on availability, suitability, and representativeness of the material. • Approve (discussion) the acquisition of raw material through the Operations Manager.	
ASMF7.2.2 Material selection record	This record ensures that every material used in RM production is **evaluated, justified, and documented** before use, to demonstrate if it is **fit for its intended purpose** (e.g.,	• Record full traceability information (origin, batch, date, sampler, method), *this can be found in the receiving book*.	**ISO 17034:2016 clause 7.2**

Procedure	Aim	Activities	International standard requirement
	homogeneity, stability, analyte level, and traceability).		
ASMF7.2.4 Calibration log	Calibration register for equipment		**ISO 17034:2016 clause 7.6**
ASMF 7.4 Material handling and storage	To maintain the **identity, quality, stability, and traceability** of all materials (raw, in-process, and final) through appropriate handling and storage procedures within the laboratory.	• Sampling shall be carried out in accordance with ISO 18283 and ISO 33405 (see doc **AS008IW**) • Reduce the bulk sample to laboratory test sample size using methods such as crushing, dividing, pulverizing as outline by AS001IW • Perform and retrain records of homogeneity tests • Store packaged material under controlled conditions (temperature, humidity) to preserve integrity (doc AS018TF)	**ISO 17034:2016 clause 7.4 ISO 17043:2023 clause 7.3.3**
ASMF 7.9 Metrological traceability	To ensure that all certified values assigned to reference materials are metrologically	○ A procedure was defined	**ISO 17034:2016 clause 7.12**

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Procedure	Aim	Activities	International standard requirement
	traceable to the international system of units (SI units).		
ASMF 7.12 Characterisation ASMF 7.13 Assignment of property values and uncertainties	To define a systematic, traceable, and statistically valid approach to assigning property values and estimating measurement uncertainty for reference materials produced	○ Maintain a characterisation report (this report records the homogeneity and stability assessments as well)	**ISO 17034:2016 clause 7.12 and 7.13 ISO 17043:2023 clause 7.4**
ASMF 7.15 Distribution and customer support	To define the process for the packaging, shipment, documentation and post-distribution support of proficiency testing samples and coal reference materials	○ Amongst other activities distribution log shall be maintained ○ Instructions to participants form AS001PT	**ISO 17043:2023 clause 7.3.4 and 7.3.5 ISO 17034:2016 clause 7.15**
ASMF 7.5.3 Surveillance process	To review if requirements of PT scheme were met	○ Maintain a record of area assessed	**ISO 17043:2023 clause 7.5.3**
ASMF7.16 Control of technical records	To ensure that all quality and technical records are authenticated, complete and traceable.	○ Maintain technical records of equipment verification, homogeneity test, stability test, etc ○ PT reports	**ISO 17034:2016 clause 7.16 and ISO 17043:2023 clause 7.5.1 and 7.4**

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Procedure	Aim	Activities	International standard requirement
ASMF7.17 Management of non-conforming work	The procedure for identifying, documenting, evaluating, correcting, and preventing recurrence of any non-conforming work	○ Record and assess the non-conforming work on ASMF8.7.3 IMPROVEMENT ACTION REQUEST form	**ISO 17034:2016 clause 7.17 and ISO 17043:2023 clause 7.5.4**
ASMF 7.18 Complaint and appeals handling procedure	To define the process for receiving, evaluating, investigating, and resolving complaints and/or appeals.	○ Maintain records of complaints, assessment and resolution	**ISO 17034:2016 clause 7.18 and ISO 17043:2023 clause 7.6 and 7.7**

3.4 Statistical design

For meaningful statistical evaluation, a minimum sample size of n = 10 is required.

Descriptive statistics is to be performed.
ASMF 7.2.2 Statistical Design procedure was developed and implemented.

Statistical treatment

- **Collection of results**: All participants submitted results on an air-dry basis. These results we subsequently converted to a dry basis to ensure comparability across laboratories.
- **Data screening and outlier assessment**: The dataset was examined for consistency and reliability. Outliers were identified using the criterion of ±3 standard deviations from the mean. Results flagged as outliers were excluded from further statistical evaluation. This step ensured that only representative data contributed to the assigned values
- **Statistical evaluation**: For each parameter, the following descriptive statistics were calculated. Mean, standard deviation, RSD, Median, Upper and lower control limits, UoM, standard error, min and max.

- Z-scores were also computed to assess individual laboratory performance relative to the assigned values. Graphical summaries, including population distributions and laboratory-specific plots, were generated to illustrate the spread and comparability of results.
- The procedures to be used to treat participant results from different measurement or test methods which are not technically equivalent, shall not be permitted by the scheme
- **Assignment of reference values**: The CRM certificate values were assigned from the **consensus means** of the participant data set, following removal of statistical outliers. This approach ensures that the reference values are derived from interlaboratory consensus, consistent with international PT practices.
- **Estimation of Measurement Uncertainty**:
  The uncertainty of the assigned reference values incorporates contributions from:
  - Homogeneity assessment
  - Stability Check
  - Characterisation Uncertainty
  - Transport effects were evaluated and found to be negligible; therefore, no additional uncertainty component was attributed to transportation.

  Compliance with SANAS R80: This process of reference value assignment ensures that the values are statistically valid, traceable, and supported by an uncertainty budget. It also aligns with the SANAS R80 requirement to clearly document the source of the assigned values and the methods used in their determination.

3.5 Homogeneity and stability assessment of PT items

3.5.1 The assessment of homogeneity and stability shall be performed for every PT round after the PT items have been packaged in their final form. The calculated results shall be included in the PT report.

3.5.2 Criteria for suitable homogeneity is established as follows:

- 10 random samples are extracted from the batch during packaging. These are used for homogeneity assessment prior to the distribution of samples to participants.
- Ash% parameter Test method is used for both purpose of homogeneity test and stability test. Reason is that Ash% parameter in coal is the most studied and stable parameter to be used for homogeneity and stability checks.
- Homogeneity is an important requirement for all PT items and RM items and includes both within- and between-unit homogeneity. The statistics is calculated using ANOVA and it is

required to achieve between-unit less than (<) the check value. It ensures that each PT item and each RM item carry the same value for each property.

- Within-unit homogeneity calculated result compared to check value is important to demonstrate that where subsample measurement is assessed against an expected certified reference value (check value). A large difference between the calculated and check value indicates a potential error in the process, the measurement or the formula used. Therefore, a decision can be made to accept the result or stop the process, recalibrate or reject the results.
- Thus this safeguards the risks that inhomogeneity and instability can impact the evaluation of the performance of participants

3.5.3 The assessment for stability shall be performed for the PT items to demonstrated sufficient stability and ensure that they will not undergo any significant change throughout the conduct of the PT round, including storage and transport. A sufficiently stable stability assessment provides confidence to the end user of the RM item to rely on the assigned value at any point within the period of validity of the certificate.

- 10 samples which were used for the homogeneity assessment are analysed to check for stability, the Ash% test method is used for this purpose for the same reason provided previously.
- The mean difference between homogeneity and stability assessments is compared. The difference shall be less than (<) the check value. Then, the PT samples meet stability requirements

3.5.4 A continuous study is performed monthly at the laboratory to determine and conclude stability of RM items under long term storage condition which is (1) one year of opening the bottle. The result will be updated accordingly at the end of the study for all RM units produced.

3.5.5 PT items from previous PT rounds, retained for another PT round, the property values or characteristics to be determined in the PT scheme shall be confirmed prior to distribution.

3.6 Instructions for participants

Distribution of PT items shall be conducted by no later than 10th of every month. Instructions to participants shall be sent together with distribution of PT items no later 10th of every month. Submission of results shall be no later than 25th of every month, however grace period is allowed until 28th of every month until 11pm and participants who couldn't submit on the 25th of every month shall be contacted to submit results.
Feedback report to participants shall be sent no later than 06th of every month

3.6.1 Document AS001PT proficiency testing instruction form or Microsoft link form provides detailed information of the handling and participation in the PT round. The instructions shall include:

- the necessity to treat PT items in the same manner as routine samples, for an example which means instruction to condition PT item for the general analysis for a defined period shall be included.
- ensure to include conditions specified in the respective test method of interest i.e. ISO or SANS or Eskom method or ASTM, etc and meet factors which can influence the measurements or tests of the PT items.
- specific and detailed instructions on the manner of recording and reporting results
- This instruction usually includes parameters such as the units of measurement e.g. "air dry base"
- specific instructions on providing details concerning the measurement or test method used by the participant, where a single specific measurement or test method is not required;
- instructions on return or forwarding of the PT items, a link shall be provided to participants by no later than 10th of every month
- information on the contact details of the PT provider for enquiries.

3.7 Determination of assigned values

ASMF 7.12 Characterisation and ASMF 7.9 Metrological Traceability shall be used for determining the assigned values for the properties or characteristics of a PT/RM item. Each parameter will have Uchar (uncertainty of characterization) that has a guard band stated as follows: The guard band for acceptability of uncertainties is that the uncertainties assigned must be less than the reproducibility limit as per the parameter.

The CRM certificate values shall be assigned from the **consensus means** of the participant data set, following removal of statistical outliers. This approach ensures that the reference values are

derived from interlaboratory consensus, consistent with international PT practices. An estimate of the uncertainty of the assigned value incorporates contributions from:

- Homogeneity assessment
- Stability Check
- Characterisation Uncertainty
- Transport effects were evaluated and found to be negligible; therefore, no additional uncertainty component was attributed to transportation.

NB: A detailed uncertainty budget is provided in document UB001 uncertainty budget (Master)

3.8 Disclosure of assigned values

In line with the requirements of ISO 17043:2023 clause 7.2.3.5, measures are in place to ensure that participants cannot gain an advantage through early disclosure of assigned values. The following practices are applied:

- Single submission: each participant is allowed to submit their results once. No edits or re-submissions are allowed.
- Post-closure analysis: All statistical evaluation and assignment of values are performed only after the submission deadline
- Simultaneous release: final PT reports, including the assigned values are issued to all participants at the same time. This ensures that no participant has access to assigned values ahead of others.

The integrity of proficiency testing scheme is maintained through these practices and participants are assured of fair and unbiased evaluation.

3.9 Evaluation and reporting of PT scheme results

Results received from participants shall be recorded and analysed using descriptive statistics. Procedures shall be established and implemented to check the validity of data entry, data transfer, statistical analysis, and reporting.
Microsoft forms links are used to collect and deposit data entry from participants, it then transfers input data into excel workbook. Data analysis shall generate summary statistics and performance statistics and associated information consistent with the statistical design.

The influence of outliers on summary statistics shall be minimized by using the criterion of ±3 standard deviations from the mean. Results flagged as outliers were excluded from further statistical evaluation. The test methods that participants are allowed to participate on are the following:

Parameter	Result (air-dry basis)	
ISO Ash (ISO 1171)		
Quick Ash		
ISO Volatile (ISO 562)		
ISO Moisture (SANS 5925)		
Calorific Value (ISO 1928)		
Total Sulphur (ASTM D4239)		
Total Moisture (semi-prep: ISO 589)		
Hardgrove Index (ASTM D409)		
Abrasive Index (Eskom method)		
Hydrogen (ISO 5074)		
Total Carbon		
Ash Fusion Temperature	*DT*	
	ST	
	HT	
	FT	

In accordance with SANAS R80, small interlaboratory comparisons are defined as those with seven or fewer participants. As this scheme is not classified as a small ILC, Azuresky applies an internal threshold of n ≥ 10 for the calculation and reporting of z-scores. This criterion is applied to ensure that statistical estimates (mean, standard deviation, and z-scores) are robust and not unduly influenced by small datasets. Where the number of participants is fewer than 10, only indicative results on an air-dry basis are provided.

The PT scheme preparation is structured in a way that no PT item goes to the participants prior to its homogeneity study being conducted and found to be sufficient. Stability checks for each PT round robin is strategically placed on the 25th of each month, a date prior to the statistical analysis of PT rounds to ensure that a stability of a sample is conducted and if any instability is identified, the PT round can be rendered null and void. This method provides a criteria and procedure to ensure that the PT items distributed and the collected data are suitable for performance evaluation.

3.10 Evaluation of performance

Interpretation of trends requires a minimum of five data points to ensure a meaningful analysis, while graphical representation necessitates at least two data points to establish a visual comparison.

When the sample size is sufficiently large, the sampling distribution of the sample mean approaches a normal distribution, regardless of the shape of the original population distribution. This assumption underpins all statistical analyses and calculations for the evaluation of performance.

The preferred method of evaluation which meet the objectives of the PT scheme. It is described as follows:
The criteria for evaluation of performance is based on consensus statistics from the current round or previous round, then a robust estimate of the standard deviation of participant results is the preferred statistic. This approach enables the performance score of z-scores and to set the standard deviation for proficiency assessment to the calculated estimate of the standard deviation.

Expert commentary shall be provided on the performance of participants with regard to the following:
a) overall performance against prior expectations, that the z-scores fall within the normal distribution curve
b) variation within and between participants, and comparisons with any previous PT rounds, or published data;
c) variation between measurement or test methods;
d) possible sources of error (with reference to outliers or poor performance) and suggestions for improving performance;
e) advice and feedback to participants as part of the continuous improvement procedures of participants;
f) situations where unusual factors make evaluation of results and commentary on performance impossible;
g) any other suggestions, recommendations or general comments;
h) conclusions.

3.11 PT reports

PT report template AS-PT001R general analysis was developed and implemented, AS-PT002R for hardgrove index was developed and implemented. Reports shall be clear, accurate, objective and comprehensive and include data covering the results of all participants, together with an indication of the performance of individual participants**.**

Reports shall include the following:

- the name and contact details (email address);

- identification of person(s) authorizing the report;
- an indication of which activities are provided by external providers when they affect the production or characterization of the PT items or the services provided;
- the date of issue and status (e.g. preliminary, interim, or final) of the report;
- unique identification that all its components are recognized as a portion of a complete report and a clear identification of the end;
- a statement of the extent to which results are confidential;
- a unique identification of the report and the PT scheme;
- a clear description of the PT items used, including necessary details of the PT item's production and homogeneity and stability assessment;
- the results of participants, including the reported measurement uncertainties (if required);
- procedures used to statistically analyse the data;
- statistical data and summaries, including assigned values, range of acceptable results and graphical displays;
- details of the metrological traceability, and uncertainty of any assigned value;
- procedures used to establish any assigned value and its uncertainty;
- assigned values, their uncertainties and summary statistics for measurement or test methods used by each group of participants (if different measurement or test methods are used by different groups of participants);
- procedures used to establish the standard deviation for proficiency assessment, or other criteria for evaluation;
- comments on the performance of participants;
- information about the design and implementation of the PT scheme;
- advice on the interpretation of the statistical analysis;
- comments or recommendations based on the outcomes of the PT round.

ASMF 7.4.3. Participant use of Proficiency Testing report policy, was developed and implemented. The laboratory‘s participation in Azuresky PT scheme will be evaluated against their plan. Failure of laboratories to show effective participation could result in suspension of accreditation. The laboratory shall make available to the assessment team all proficiency testing scheme reports.

When it is necessary to issue a new or amended report for a PT round, this report shall include the following:
a) a unique identification;
b) a reference to the original report that it replaces or amends;
c) identification of the amendment and a statement concerning the reason for the amendment or re-issue.

It is important to ensure that when issuing an amended report to a subset of participants, an analysis of the potential impact on the other participants for that PT round shall be made to ensure there is no influence on the general performance of the other participants.

If there is a need to issue a statement of participation or performance in addition to the PT report, the statement shall not be misleading.

3.12 Surveillance of the processes

Azuresky is committed to ensure the validity of the PT scheme. This is to gain benefit of information on the comparability of test or measurement methods, benchmarking against other PT scheme or RM producers, etc

Surveillance activities shall be planned and reviewed [see also 8.9.2 item n)], and the resulting data shall be recorded for the continuous improvement process.
NOTE Depending on the PT scheme, surveillance activities can include:

— use of reference materials or other control items;
— the transmission of results from participants;
— control of statistical conditions to confirm the validity of performance evaluation;
— checking of reports;
— for continuous schemes, comparisons against previous PT rounds.

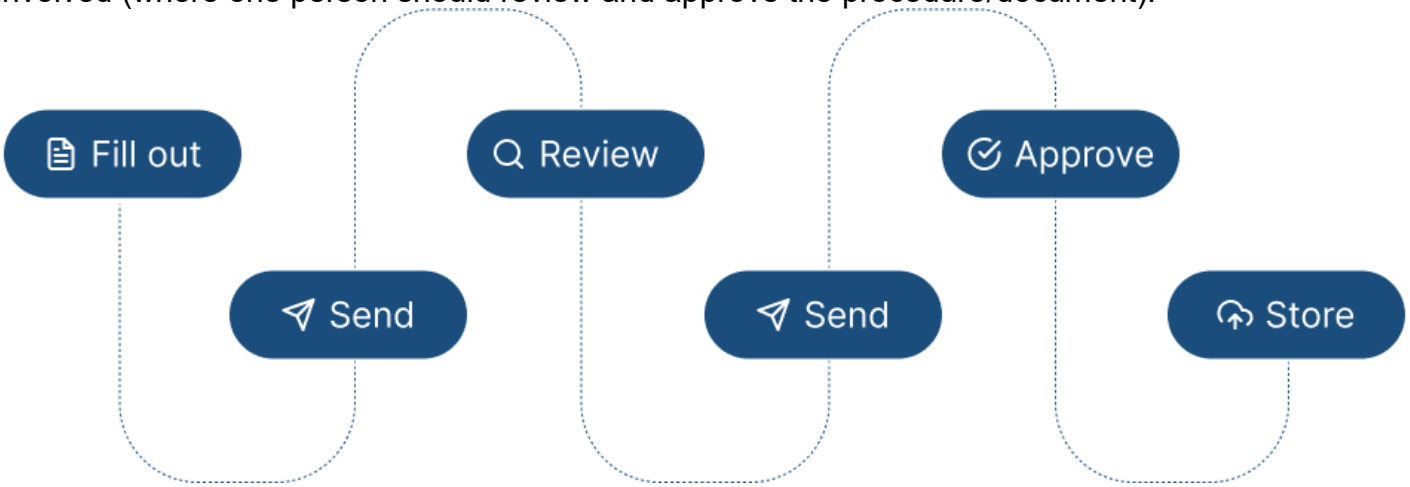
	Azuresky	
	Management system requirements	Document no: ASDP001
	Compiled by:	**Sifiso Nyambi - Laboratory Manager**
	Approved by:	**Nonhlanhla Msibi - Operations Manager**

4. Management System Requirements

4.1 Control of management system documents

ISO 17043:2023 clause 8.3; ISO 17025:2017 clause 8.3 and ISO 17034 clause 8.4: Document control is a requirement for all companies and must be in place to manage document creation, review, approval, distribution, and archiving.
This helps ensure that the documentation is controlled, consistently maintained, and organized. A formal document review and approval process shall be utilized to ensure accuracy and compliance with regulatory requirements.

The process flow illustrated on diagram below may be followed by the team members. Where three persons may contribute to the development of the procedure/document or two persons involved (where one person should review and approve the procedure/document).



4.1.1 Implement Version Control

Version control has been implemented to track and manage document revisions. Procedures have revision table which has ensured to indicate the document version number, the content of the change, who modified the document, the reason for change and date and time to avoid confusion. The latest version shall be readily identifiable and accessible. The old document shall be retired (file in obsolete folder) to avoid the use of outdated documents. Communication of changes to relevant persons shall be conducted to ensure adequate implementation. Evidence of communication shall be retained.

Effective date	Rev status	Next review date
04/2025	00	04/2028

It is imperative that comprehensive training plans must be conducted to educate employees on the purpose, content, and use of documentation. A skill matrix have been developed to assess employees' competencies, identify skill gaps, and support the creation of training plans.
By providing effective training, the organisation equip employees with the necessary skills and knowledge to create, manage, and utilize documentation effectively throughout their roles.

4.1.2 Ensure Document Accessibility

ISO 17025 clause 8.2.5 and ISO 17043 clause 8.2.3 states that all personnel involved in laboratory/PT activities shall have access to the parts of the management system documentation and related information that are applicable to their responsibilities.
Only authorized personnel (Azuresky team members) have access to the required documentation. The storage is managed through QMWARELIMS12 in Azuresky system folder. All team members have access to the folder. However restrictions to access have been applied to certain sub-folders.
For example: Subfolder for HR, access is limited and not accessible to all team members.
Other hardcopy management system documents eg. Calibration certificates are stored in files and accessible to all personnel related to their responsibilities.

4.2 Control of records

The records produced by the laboratory shall be retained for 10years minimum and/ or the contractual obligation terms shall be superseding the 10years period if it is greater. A record matrix form ASCR 001, lists the records produced by the laboratory, storage, protection, back-up, archive, retrieval and disposal.

The following conditions are required to control the records:

- Technical records for each PT/RM activity contain the results, reports and sufficient information to facilitate, if possible, identification of factors affecting the PT/RM performance evaluation and its associated characteristics and enable the repetition of the PT/RM activity under conditions as close as possible to the original. The technical records shall include the date and the identity of personnel responsible for each PT/RM activity and for checking data and results.
- Data used to verify the PT/RM items, instructions to participants, the original responses of participants and any other information included in reports shall be recorded at the time they are made and shall be identifiable with the specific task.
- **A**mendments to technical records can be tracked to previous versions or to original information submitted by participants. Both the original and amended data and files shall

be retained, including the date of alteration, an indication of the altered aspects and the personnel responsible for the alterations.

4.2 Control of data and information management

- All relevant personnel as per job description shall have access to the data and information needed to perform their activities.
- **QMWARE version 12** information management system is used for the collection, processing, recording, reporting, storage or retrieval of data. It shall be validated for functionality, including the proper functioning of interfaces before introduction. Whenever there are any changes, e.g. software configuration, etc, they shall be authorized, documented and validated before implementation.
- **QMWARE version 12** information management system shall:
  a) be protected from unauthorized access;
  b) be safeguarded against tampering and loss;
  c) be operated in an environment that complies with the system supplier, input of data into LIMS is not auto transfer from measuring equipment. Every mass input is captured in LIMS and the operator involved as well. Changes to any mass inputs and results are also captured automatically in LIMS. Such changes are password protected and privileges are only given to management to make such changes in the case an error that may arise during analysis. If any change occurs in a spreadsheet, LIMS automatically flags this spreadsheet each time it is opened.
  d)be maintained in a manner that ensures the integrity of the data and information;
  e) include recording of system failures and the appropriate immediate and corrective actions.
- **QMWARE version 12** information management system shall ensure tutorial videos which serve as instructions, manuals and reference data relevant to the laboratory information management system are made readily available to personnel.
- Calculations and data transfers shall be checked.

4.3 Actions to address risks and opportunities

A risk assessment to identify risks and opportunities associated with the PT and RM production is to be undertaken annually by all personnel involved in the running of the PT scheme and RM production.
When there are plans to introduce significant changes to activities which can affect the validity of the PT scheme, risk assessment shall be performed to identify and manage the risk to ensure the validity of the PT scheme is maintained. Hence this will ensure the validity of RM production.

A document titled 8.5 Actions to address risks and opportunities (excel sheet) is to be completed for this purpose and actions arising from the exercise are to be planned and executed accordingly.

4.4 Improvement

It is in the best interest of Azuresky (Pty) Ltd to improve its processes and delivery of PT scheme and RM production so to capture the marketplace. Amongst other benefits is to seek to increased efficiency, improved customer satisfaction, and better regulatory compliance. Therefore the guideline of ISO 17043:2023 clause 8.6; ISO 17025:2017 clause 8.6 and ISO 17034:2016 clause 8.8 shall be performed.
Opportunities arising from but not limited to the following list shall be performed.

- Management review meeting which is to be held annually (January month).
- Company objectives
- Audit results
- Suggestions from personnel
- Proficiency testing results
- Review of instructions to work/operational procedures
- Corrective actions
- Risk assessment
- Analysis of data
- External assessment eg. Customer feedback form AS006MF was developed and implemented for use to engage with clients and gather feedback of the service.

Actions shall be planned, implemented and monitored. Feedback from external assessment shall be analysed and used to improve the management system, PT and RM activities and customer service.

4.5 Corrective Actions

An improvement form document no. ASMF 8.7.3 was developed and implemented to record the non-conformities identified. If necessary, risks and opportunities will be identified, assessed, implemented and updated.
The storage of records is managed through QMWARELIMS12 in Azuresky system folder for softcopy and hardcopy are stored in files.
Corrective actions shall form item of discussion during the management review meeting and establish improvement and effectiveness of the corrective actions implemented.

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4.6 Internal audits

Internal audits plan AS005MF was developed and implemented. It shall be updated as necessary in terms of scheduling.
The criteria for an independent person to conduct an audit, a competent experienced person in the field will be consulted to conduct the internal audit however this will be dependent on the availability of funds. Operations Manager will conduct the audit in case of financial constraint.

When audit findings cast doubt on the effectiveness of the operations, or on the integrity of the PT or RM samples or the correctness of their documentation, Azuresky Management shall timely correct the deviations and shall notify in writing the customers whose activities may have been adversely affected.

4.7 Management reviews

Management review meetings shall be conducted annually in January month. The inputs as described in ISO 17043:2023 clause 8.9.2; ISO 17034:2016 clause 8.6.1 and ISO 17025:2017 clause 8.9.2 shall be discussed.

The purpose of management review is to ensure continuing suitability and effectiveness, and introduce necessary changes and improvements

A record of the minutes of the meeting shall be developed, reviewed within two weeks of the meeting and signed by the Operations Manager and the Laboratory Manager. This is to confirm the contents of the minutes as true facts of discussion. Then minutes shall be shared immediately with all team members (all employees).